Mall Customer Segmentation & EDA Report

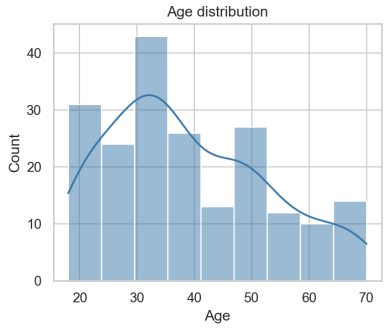
Comprehensive Machine Learning & Unsupervised Clustering Analysis | Dataset: Mall_Customers.csv

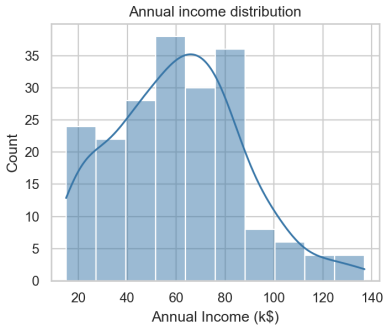
1. Executive Summary & Data Quality Assurance

- **Data Ingestion:** 200 customer profiles with 5 attributes: CustomerID, Gender, Age, Annual Income (k$), and Spending Score (1-100).
- **Data Cleaning:** Verified zero missing values, zero duplicate CustomerIDs, trimmed whitespace, and validated valid numeric ranges.
- **Standardization:** Applied Z-score standardization (StandardScaler) to Annual Income and Spending Score to eliminate scale bias.

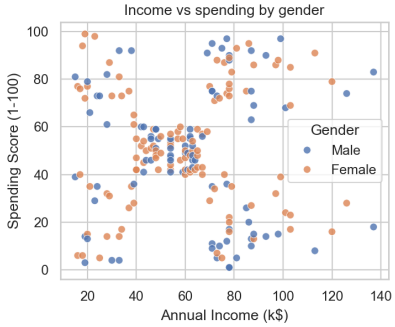
2. Exploratory Data Analysis (EDA)

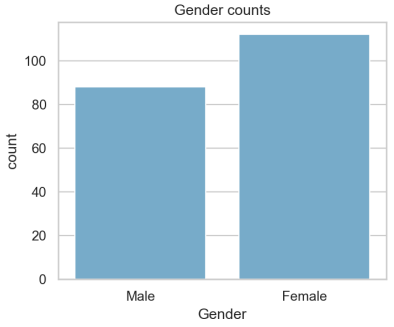
Comprehensive distributions, correlations, and cross-feature interactions across the customer base:

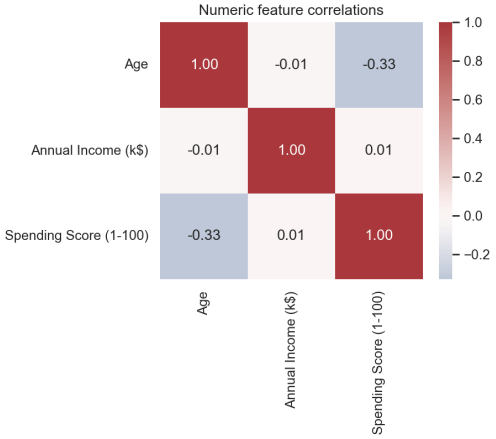










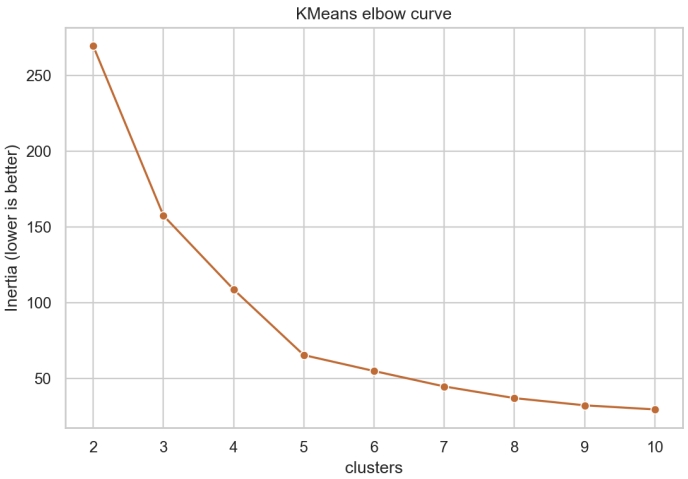


3. Unsupervised Model Benchmarking & Metric Comparison

Evaluated KMeans, Agglomerative Hierarchical Clustering (Ward linkage), Gaussian Mixture Models (GMM), and DBSCAN across clusters k=2..10:

Model Architecture	k	Silhouette Score (↑)	Davies-Bouldin (↓)	Calinski-Harabasz (↑)
KMeans (k=5)	5	0.5547	0.5722	248.65
Agglomerative Ward (k=5)	5	0.5538	0.5779	244.41
Gaussian Mixture (k=5)	5	0.5537	0.5760	244.94
KMeans (k=6)	6	0.5399	0.6546	243.09
Agglomerative Ward (k=6)	6	0.5387	0.6449	233.31

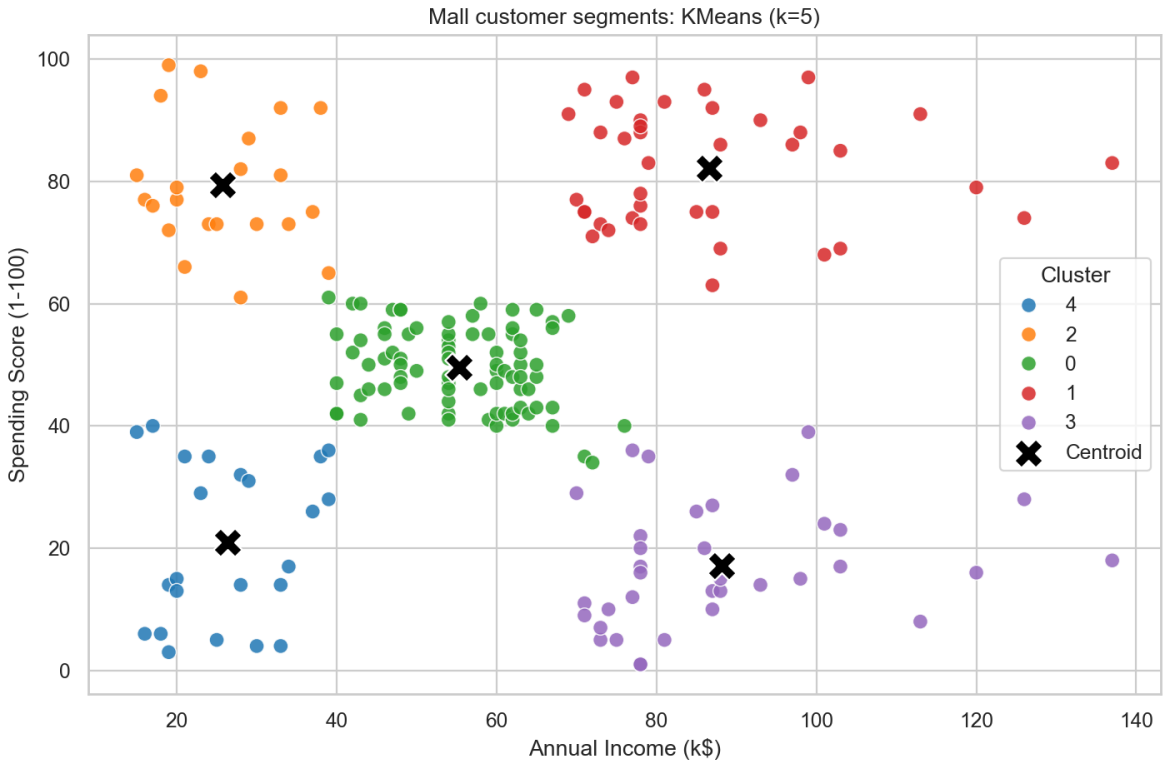




4. Selected Segmentation: KMeans (k=5)

KMeans (k=5) achieved the highest Silhouette Score (0.5547) and lowest Davies-Bouldin Index (0.5722), matching the elbow curve inflection point.

Cluster	Size	Avg Age	Avg Income	Avg Spend Score	Customer Persona & Marketing Strategy
Cluster 0	81	42.7 yrs	$55.3k	49.5/100	Middle-of-the-Road: Moderate income & spend. Offer balanced seasonal promotions.
Cluster 1	39	32.7 yrs	$86.5k	82.1/100	Target VIP High-Spenders: High income & high spend. Premium luxury loyalty perks.
Cluster 2	22	25.3 yrs	$25.7k	79.4/100	Careless Youth Spenders: Low income, high spend. Trend merchandise & viral campaigns.
Cluster 3	35	41.1 yrs	$88.2k	17.1/100	Frugal Savers: High income, low spend. High-value propositions & financial utility.
Cluster 4	23	45.2 yrs	$26.3k	20.9/100	Budget-Conscious: Low income, low spend. Deep discounts, clearance sales, coupons.


Mall customer segments: KMeans (k=5)
Spending Score (1-100)
Annual Income (k$)
0
20
40
60
80
100
120
140
Cluster
4
2
0
1
3
Centroid